



BKP: An R Package for Beta Kernel Process Modeling

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joint work with Kunhai Qing and Jin Xu

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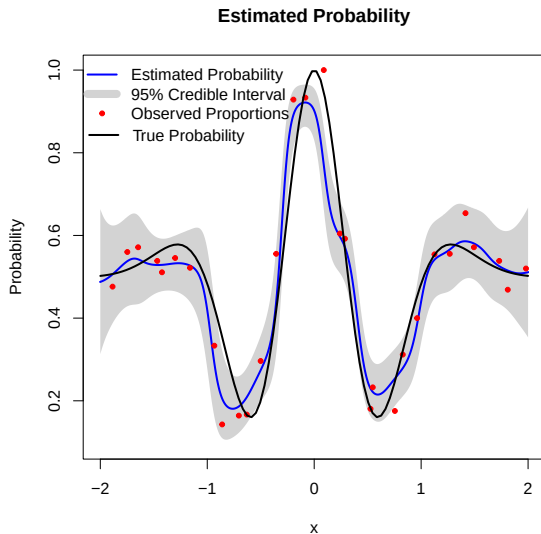


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Estimating an Unknown Probability Surface



- **Observation model**

$$y_i \mid \pi(\mathbf{x}_i) \sim \text{Binomial}(m_i, \pi(\mathbf{x}_i)).$$

- **Objective**

Estimate the underlying probability surface $\pi(\mathbf{x})$ and quantify its pointwise uncertainty.



Existing Approaches and the Remaining Gap

Logistic regression

$$\pi(\mathbf{x}) = \text{logit}^{-1}(\mathbf{x}^\top \boldsymbol{\beta})$$

Fast and interpretable
Limited nonlinear flexibility

Gaussian process classification

$$\pi(\mathbf{x}) = \text{logit}^{-1}\{f(\mathbf{x})\}, \quad f(\mathbf{x}) \sim \text{GP}$$

Flexible probability surfaces
Approximate posterior inference

Motivation

Can we retain nonlinear flexibility while performing direct, closed-form updating on the probability scale?



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Beta Kernel Process: Direct Probability-Scale Updating

Beta prior

$$\pi(\mathbf{x}) \sim \text{Beta}\{\alpha_0(\mathbf{x}), \beta_0(\mathbf{x})\}.$$

Prior information on the probability scale

+

Local likelihood

$$k_i(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}_i; \boldsymbol{\theta}),$$

$$S(\mathbf{x}) = \sum_{i=1}^n k_i(\mathbf{x}) y_i,$$

$$F(\mathbf{x}) = \sum_{i=1}^n k_i(\mathbf{x}) (m_i - y_i).$$

Nearby observations receive greater weight

$\Rightarrow$

Beta posterior

$$\pi(\mathbf{x}) | \mathcal{D} \sim \text{Beta}\{\alpha_n(\mathbf{x}), \beta_n(\mathbf{x})\},$$

$$\alpha_n(\mathbf{x}) = \alpha_0(\mathbf{x}) + S(\mathbf{x}),$$

$$\beta_n(\mathbf{x}) = \beta_0(\mathbf{x}) + F(\mathbf{x}).$$

Closed-form posterior inference

Key idea

Local kernel weighting + Beta-binomial conjugacy
 $\Rightarrow$ **flexible modeling with closed-form posterior inference.**



Closed-Form Posterior Summaries

Probability estimation

$$\hat{\pi}(\mathbf{x}) = \frac{\alpha_n(\mathbf{x})}{\alpha_n(\mathbf{x}) + \beta_n(\mathbf{x})}$$

Posterior mean of the unknown probability surface

Pointwise uncertainty

$$\text{Var}\{\pi(\mathbf{x}) \mid \mathcal{D}\} = \frac{\hat{\pi}(\mathbf{x})\{1 - \hat{\pi}(\mathbf{x})\}}{\alpha_n(\mathbf{x}) + \beta_n(\mathbf{x}) + 1}$$

$$\text{CI}_{1-\delta}(\mathbf{x}) = [Q_{\delta/2}(\mathbf{x}), Q_{1-\delta/2}(\mathbf{x})]$$

Credible intervals follow from Beta posterior quantiles

Count prediction

$$Y_{\text{new}} \mid \mathcal{D}, \mathbf{x}, m_{\text{new}} \sim \text{Beta-Binomial}\{m_{\text{new}}, \alpha_n(\mathbf{x}), \beta_n(\mathbf{x})\}.$$

Closed-form prediction for future binomial counts

Computational implication

Conditional on the kernel hyperparameters, posterior summaries and predictive distributions are available without numerical posterior approximation.



Prior Specification

Non-informative prior

$$\pi(\mathbf{x}) \sim \text{Beta}(1, 1).$$

Default uniform prior when no external information is available

Informative prior with fixed mean

$$\alpha_0(\mathbf{x}) = r_0 p_0, \quad \beta_0(\mathbf{x}) = r_0(1 - p_0)$$

$$\mathbb{E}\{\pi(\mathbf{x})\} = p_0, \quad \alpha_0(\mathbf{x}) + \beta_0(\mathbf{x}) = r_0$$

A fixed prior mean p_0 with precision controlled by r_0

Data-adaptive prior

$$p(\mathbf{x}) = \sum_{i=1}^n \frac{k(\mathbf{x}, \mathbf{x}_i)}{\sum_{i=1}^n k(\mathbf{x}, \mathbf{x}_i)} \frac{y_i}{m_i}, \quad r(\mathbf{x}) = r_0 \sum_{i=1}^n k(\mathbf{x}, \mathbf{x}_i)$$

$$\alpha_0(\mathbf{x}) = r(\mathbf{x})p(\mathbf{x}), \quad \beta_0(\mathbf{x}) = r(\mathbf{x})\{1 - p(\mathbf{x})\}.$$

The prior mean follows the local empirical proportion, while the prior precision adapts to the local kernel mass



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Summary

- Developed the **first publicly available BKP package** implementing BKP and DKP models
- **Methodological contributions:**
 - Provided a **Bayesian interpretation** of the BKP model
 - Proposed a **flexible framework for constructing priors**, including data-adaptive options
 - Developed a **robust and efficient hyperparameter optimization strategy** using LOOCV and multi-start reparameterization.



Future Development Directions

- **Model extensions:** Extend the BKP framework to multivariate, functional, and spatio-temporal data, as well as mixed-type covariates
- **Alternative likelihoods:** Explore likelihoods beyond the binomial family, e.g., negative binomial for over-dispersed counts, geometric for waiting-time modeling
- **Community contributions:** We invite developers to contribute via GitHub pull requests <https://github.com/Jiangyan-Zhao/BKP/pulls>



Resources

-  <https://cran.r-project.org/web/packages/BKP/index.html>
-  <https://github.com/Jiangyan-Zhao/BKP>
-  <https://arxiv.org/abs/2508.10447>



